

Fig. 3: Typical movement of liquid through a peristaltic pump

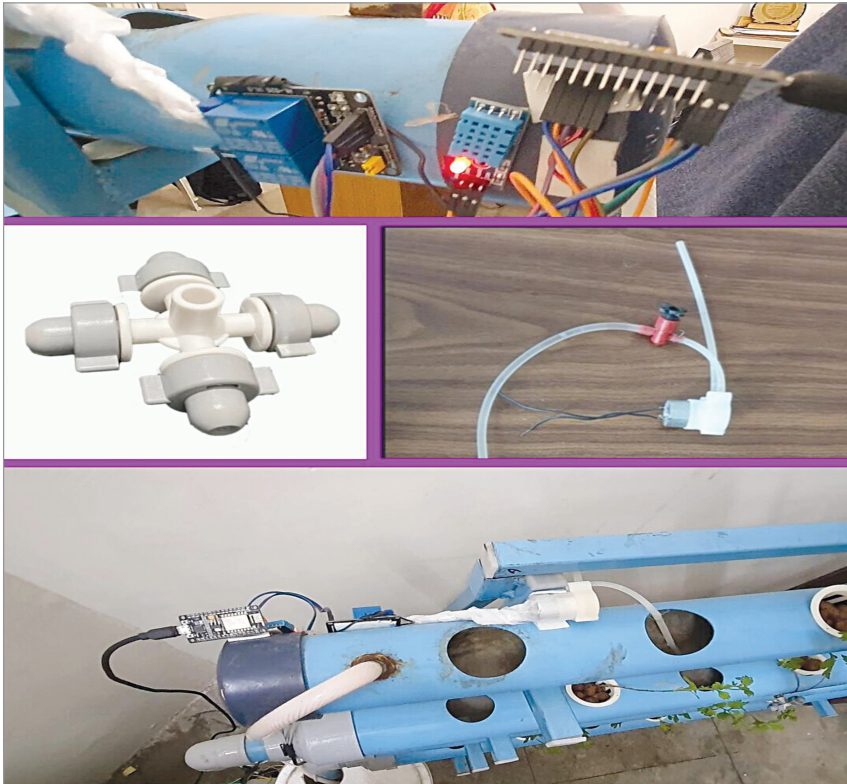


Fig. 4: The modified setup that fulfills the demand for the desired flow

TABLE 1
BILL OF MATERIAL

Components	Amount	Description
ESP8266 (MOD1)	1	For programming
DHT11 (S1)	1	To sense atmospheric temperature and humidity
DS18B20 (S2)	1	To sense the water temperature
Peristaltic pump (P1)	1	To supply water to the fogger
Dual channel relay (MOD 3)	1	To control the peristaltic pump
Water flow valve (MOD 2)	1	To control water flow
Misting nozzle (N1)	1	To mistify the water for the water shower
6.4mm (¼-inch) barbed connector	1	To connect microtubes
6.4mm (¼-inch) microtube	¾ metre	To connect peristaltic pump with the misting nozzle and water supply
Jumper wires (female to female)	9	To connect sensors and relay

on-off controls for the fogger using additional valves and a peristaltic pump for variable flow control. Fig. 1 showcases the author's prototype at different stages. The fogger on the left-hand side represents the commercially available product, while the modified set is displayed on the right-hand side. Fig. 2 presents the block diagram representation of the entire setup.

A peristaltic pump works by using either two shoes (for high-pressure applications) or rollers (for low-pressure applications) that rotate on a wheel inside the pump. This rotation compresses an internal hose, forcing a known quantity of fluid through it, hence the name of the product. Peristaltic pumps are commonly used to move fluids that are sterile or contain aggressive chemicals. They can also pump IV fluids or slurries with a high fluid content, and they find application in heart/lung machines during surgery. Peristaltic pumps deliver pulsatile flow, and the frequency and amplitude of the required oscillations to obtain a given flow rate vary depending on factors such as the number of rollers, their rotation frequency, and the diameter of the tubes. The mechanical peristaltic operation involves rotating rollers that depress the silicone tubing, creating a vacuum, which displaces any fluid or gas in the chosen direction. This operation is effective up to the suction lift limit, which varies with elevation. Fig. 3 demonstrates the typical movement of liquid through a peristaltic pump.

Peristaltic pumps are an excellent option for fluids that need to be kept clean and sterile, such as in food and drink applications, as they prevent contamination from occurring in the pump. The interior tubing is the only part of the pump that comes into contact with the pumped application, minimising the risk of bacterial growth.

Peristaltic pumps offer numerous benefits. They are fully self-priming up to 9.8 metres, and they operate